



**MERU UNIVERSITY
OF SCIENCE & TECHNOLOGY**

***IOT-BASED AIR QUALITY MONITORING SYSTEM
PROPOSAL PRESENTATION***

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EG209/109705/22

BTech. Electrical and Electronic Engineering



BACKGROUND OF THE STUDY



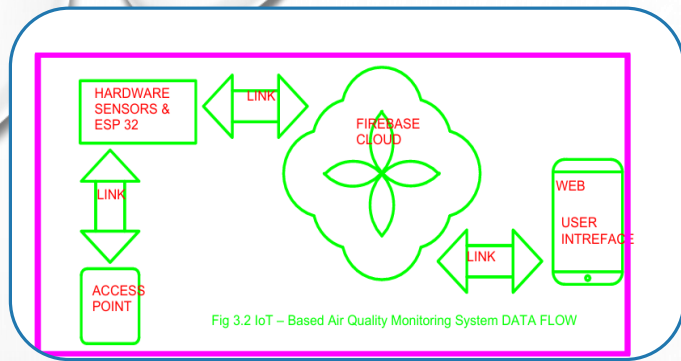
- ❖ Air pollution is a growing public health and environmental concern worldwide.
- ❖ Urbanization and industrial activities increase emission of harmful gases and particulate matter.
- ❖ Pollutants such as CO, LPG, PM2.5 and PM10 cause serious health problems.
- ❖ Gas leaks in domestic environments pose risks of fire, explosions, and poisoning.
- ❖ Existing air quality monitoring systems are centralized and expensive.
- ❖ Households lack access to real-time, localized air quality information.
- ❖ Advances in IoT enable low-cost, real-time monitoring solutions.
- ❖ This project proposes an IoT-based system for continuous monitoring and early warning.



PROBLEM STATEMENT



- ❖ Air pollution and gas leaks often go undetected in homes and small premises.
- ❖ Existing air quality monitoring systems are expensive and inaccessible to individuals.
- ❖ Lack of real-time, localized monitoring increases health and safety risks.



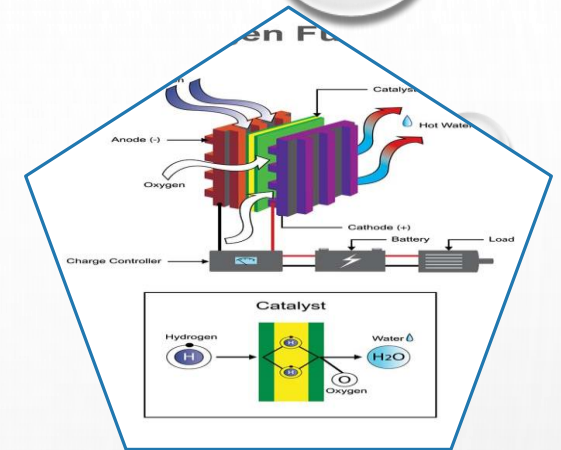
OBJECTIVES

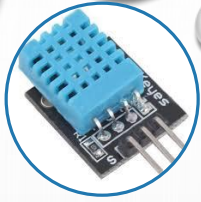
- **MAIN OBJECTIVE**

- TO DESIGN A SCALABLE IOT-BASED AIR QUALITY MONITORING SYSTEM WITH REAL-TIME DETECTION, PREDICTION, ALERTS, AND ACTUATION.

- **SPECIFIC OBJECTIVES**

1. MONITOR PARTICULATE MATTER AND HAZARDOUS GASES IN REAL TIME.
2. MEASURE TEMPERATURE AND HUMIDITY.
3. ENABLE REMOTE MONITORING VIA A CLOUD-BASED WEB APPLICATION.
4. PROVIDE INSTANT ALERTS WHEN UNSAFE LEVELS ARE DETECTED.
5. IMPLEMENT A SAFETY RESPONSE SYSTEM.





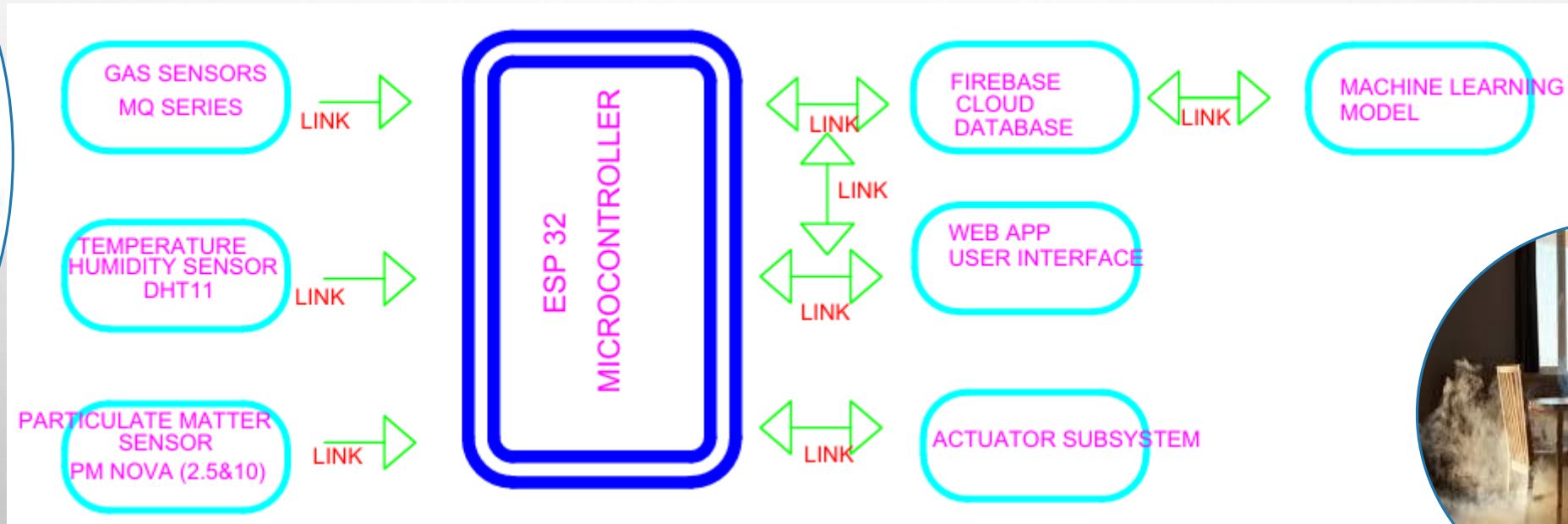
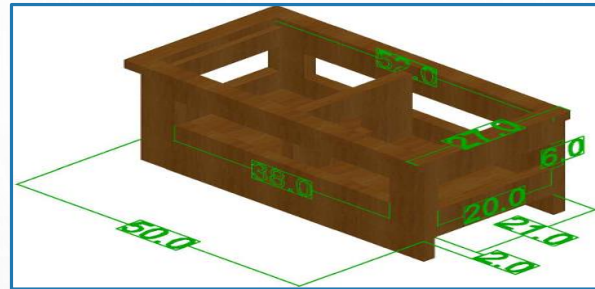
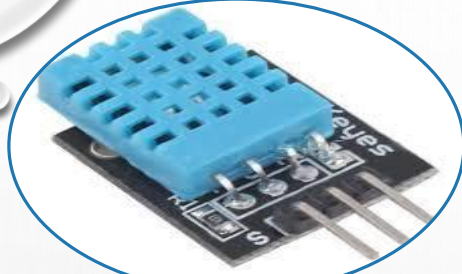
LITERATURE REVIEW



Author & Year	Focus Area	Key Findings	Identified Gap
Bagkis et al. (2025)	Low-cost sensor networks	Scalable but accuracy issues	Limited household deployment
Concas et al. (2021)	Sensor calibration	Accuracy affected by drift	Need ML calibration
Garcia et al. (2025)	IoT & AI monitoring	Real-time analytics possible	Limited safety response
Gatari et al. (2018)	Kenyan PM monitoring	Reliable low-cost sensors	Sparse local coverage
Ramesh et al. (2024)	IoT monitoring	Real-time data & alerts	Limited prediction capability



METHODOLOGY



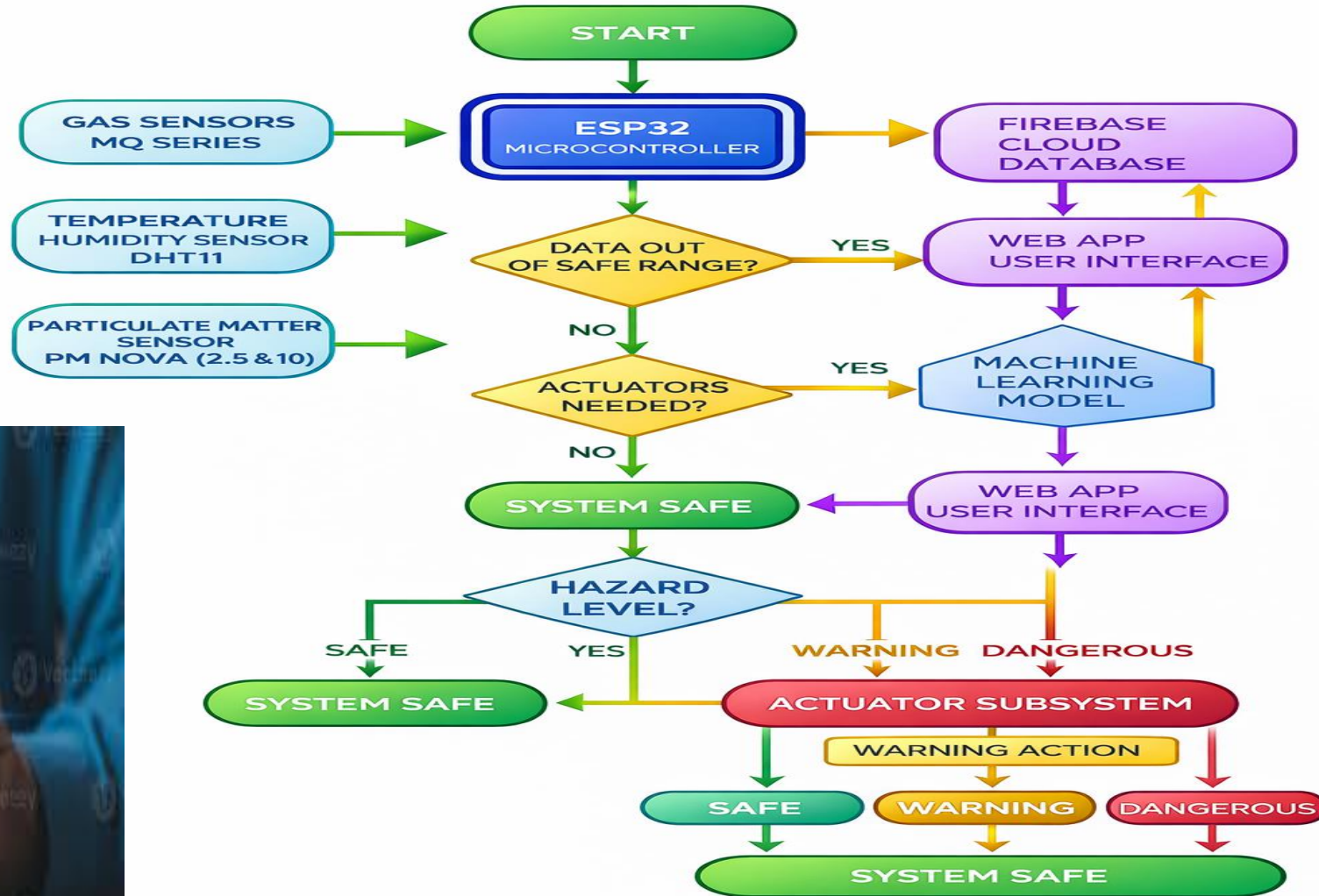


METHODOLOGY-SYSTEM OPERATION

- ❖ SENSORS CONTINUOUSLY MEASURE AIR POLLUTANTS AND ENVIRONMENTAL PARAMETERS.
- ❖ ESP32 PROCESSES AND EVALUATES SENSOR DATA LOCALLY.
- ❖ UNSAFE CONDITIONS TRIGGER IMMEDIATE ALERTS AND SAFETY ACTIONS.
- ❖ DATA IS TRANSMITTED TO FIREBASE VIA WI-FI.
- ❖ WEB APPLICATION DISPLAYS REAL-TIME AND HISTORICAL DATA.
- ❖ MACHINE LEARNING MODELS PREDICT TRENDS AND DETECT ANOMALIES.



METHODOLOGY-LOGICAL FLOW CHART





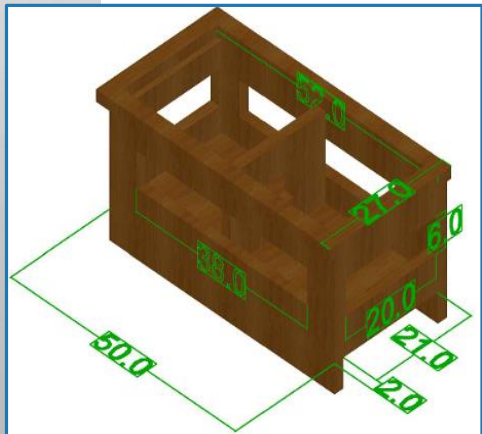
EXPECTED RESULTS



- ❖ Real-time monitoring of air pollutants and hazardous gases.
- ❖ Early detection of gas leaks and fire risks.
- ❖ Automated alerts and notifications.
- ❖ Accessible web-based visualization.
- ❖ Improved environmental awareness and safety.



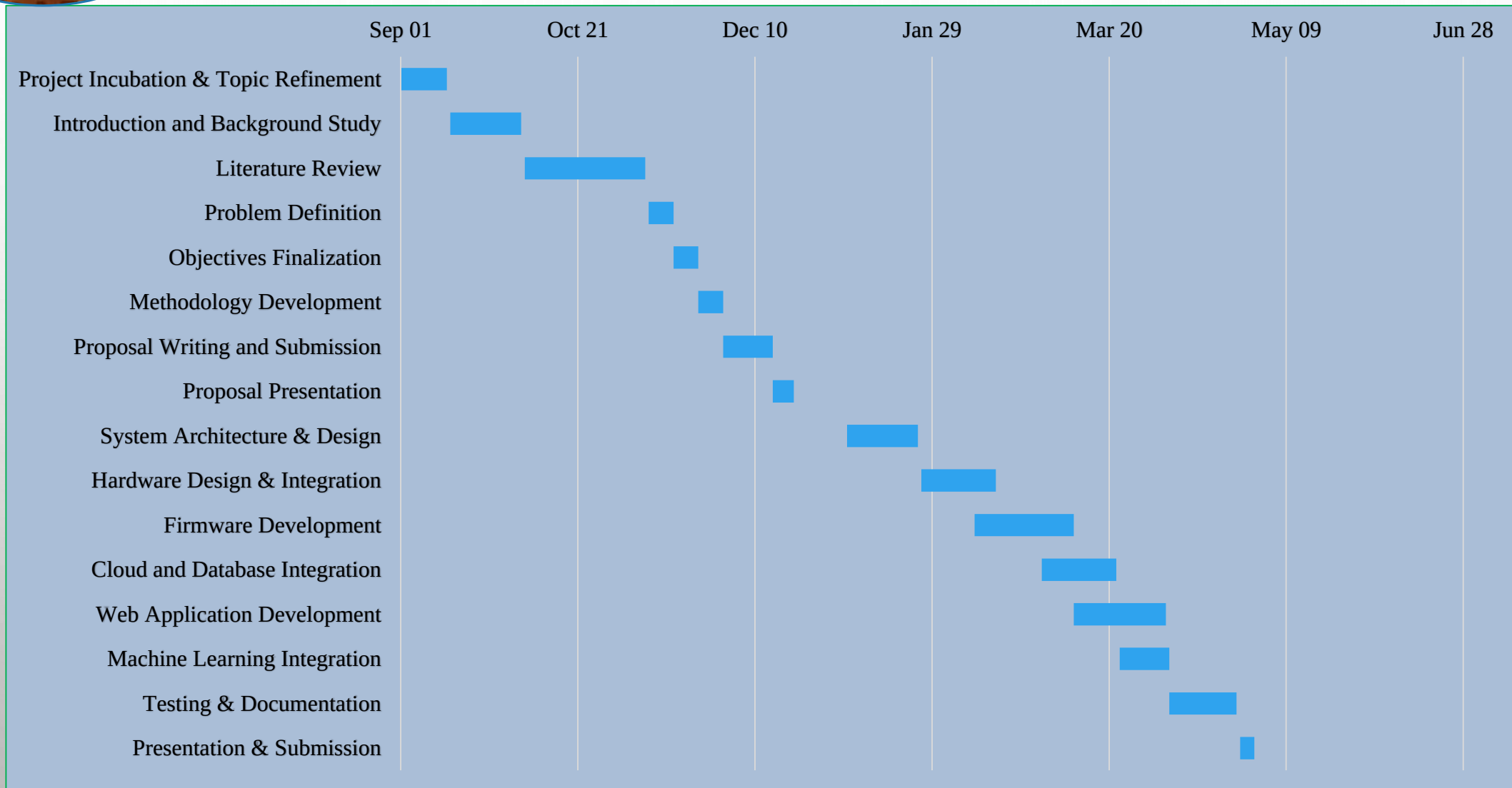
WORK PLAN



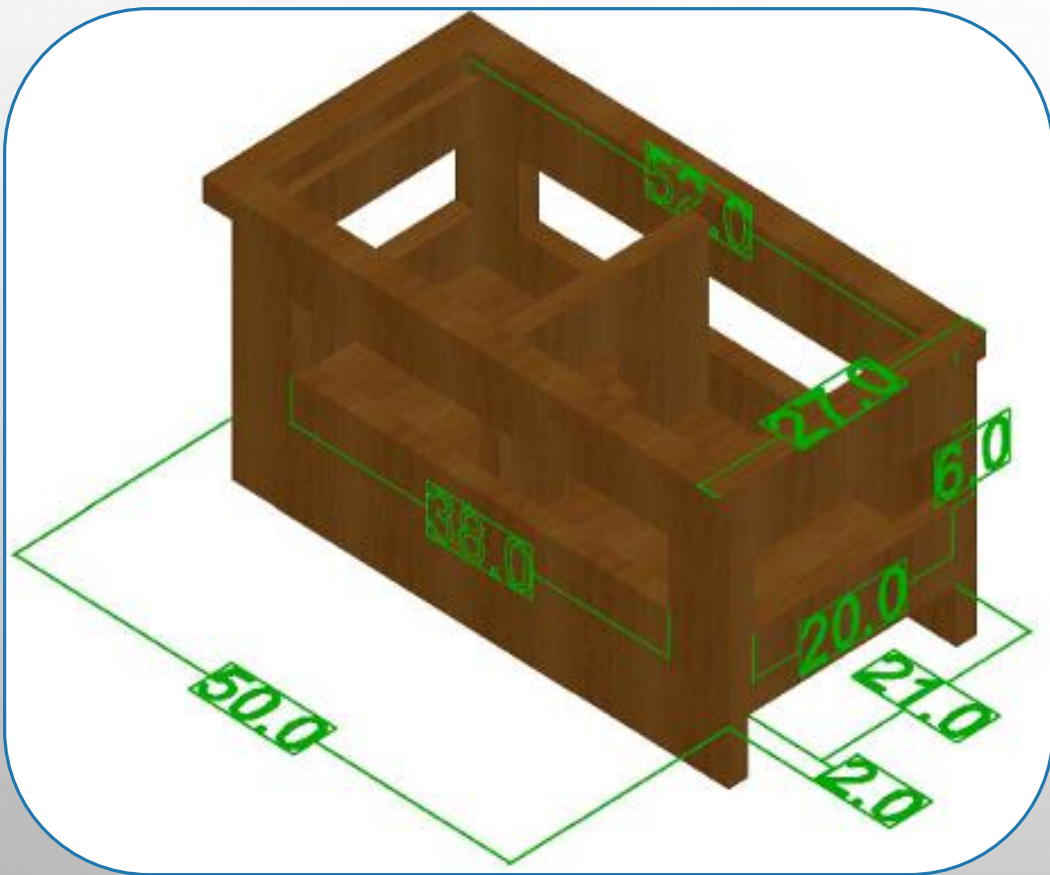
Activity	Start Date	End Date	Days
Project Incubation & Topic Refinement	01-Sep-2025	14-Sep-2025	13
Introduction and Background Study	15-Sep-2025	05-Oct-2025	20
Literature Review	06-Oct-2025	09-Nov-2025	34
Problem Definition	10-Nov-2025	16-Nov-2025	7
Objectives Finalization	17-Nov-2025	23-Nov-2025	7
Methodology Development	24-Nov-2025	30-Nov-2025	7
Proposal Writing and Submission	01-Dec-2025	14-Dec-2025	14
Proposal Presentation	15-Dec-2025	21-Dec-2025	6
System Architecture & Design	05-Jan-2026	25-Jan-2026	20
Hardware Design & Integration	26-Jan-2026	16-Feb-2026	21
Firmware Development	10-Feb-2026	10-Mar-2026	28
Cloud and Database Integration	01-Mar-2026	22-Mar-2026	21
Web Application Development	10-Mar-2026	05-Apr-2026	26
Machine Learning Integration	23-Mar-2026	05-Apr-2026	14
Testing & Documentation	06-Apr-2026	25-Apr-2026	19
Presentation & Submission	26-Apr-2026	30-Apr-2026	4



GANTT CHART



BUDGET



COMPONENTS	COST
SENSORS	4400
MICROCONTROLLER	1500
POWER SUPPLY UNIT	5900
Miscellaneous	560
CONNECTIVITY	4500
ENCLOSURE	1000
TOTAL	17860



Serial No	Component	DESCRIPTION	Unit cost	Quantity	Total
	1 ESP 32	DOIT ESP32 DEVKIT V1	1500	1	1500
	2MQ-SERIES	MQ-135 / MQ-7 gas sensors (analog modules)	400	4	1600
	3NOVA PM	SDS011 PM2.5/PM10 particulate sensor (UART)	2300	1	2300
	4BATTERY	12 V Li-ion / lead-acid battery, 2–5 Ah typical	1800	1	1800
	BATTERY				
	5CHARGER	TP4056 Li-ion charger module	1200	1	1200
	6PSU	AMS1117-3.3 voltage regulator (3.3 V output).	400	1	400
		20 cm male-to-male jumper wires. (common			
	7JUMPERS	prototyping set)	250	2	500
	8STRIP BOARD	Standard perfboard, proto assembly	100	2	200
	9BREAD BOARD	Standard 830-tie point breadboard.	200	1	200
	10ENCLOSURE	wooden box	1000	1	1000
	11INTERNET	Wi-Fi data connection plan (monthly)	1000	2	2000
	12DHT11	Temp/humidity sensor module (5 V)	500	1	500
	13ACCESS POINT	Wi-Fi router	2500	1	2500
	14POWERBANK	10 000 mAh USB power bank	1200	1	1200
	15Miscellaneous	Screws, headers, mounts.	400	1	400
	16LEDs	5 mm LEDs (e.g., red, green, blue).	10	10	100
	17BUZZER	5 V active piezo buzzer.	2	20	40
	18PUSH BUTTON	Standard tactile push button/switch	1	20	20
	19RESISTORS	Assorted: 10 k Ω , 4.7 k Ω , 1 k Ω , 220 Ω	10	20	200
	20CAPACITORS	Decoupling and bulk: 100 μ F, 10 μ F, 0.1 μ F	20	10	200
				TOTAL	17860

KEY REFERENCES



- ❖ Bagkis et al., *npj Climate and Atmospheric Science*, 2025.
- ❖ Concas et al., *ACM TOSN*, 2021.
- ❖ Garcia et al., *Artificial Intelligence Review*, 2025.
- ❖ Gatari et al., *Atmospheric Chemistry and Physics*, 2018.
- ❖ Ramesh et al., *ICCES*, 2024.
- ❖ Tastan, *Sensors*, 2025.





Thank You

